

RAILWAY TRAFFIC ANALYSIS

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MVP PRESENTATION

Client Intro - SYNOPSIS

- Synoptix is a Bristol-based engineering consultancy with expertise in systems engineering development and innovation.
- They provide expert solutions across diverse sectors, including aerospace, defence, communications, energy, healthcare, maritime, space, and transportation.

Our Project

- Modelling of railway data in order to gain a deeper understanding of the variables that affect the probability of railway incidents.
 - Incidents include: stoppages, derailments, and collisions
 - Variables include: weather conditions, temperature, track type, train type etc.
- Allows Synoptix to develop technologies to mitigate these variables.

Synoptix

As **Synoptix**, we want to identify variables that increase railway incident probability, so that we can **develop targeted, data-driven technologies**, as well as **demonstrate and justify their value and effectiveness to stakeholders**.



Railway Companies

As a **railway company**, we want to **reduce incidents** and associated **costs**, such as rail repairs, train repairs, and insurance expenses, so that we can **prioritise safety**, **prevent casualties**, **build public trust**, and maintain a reliable, **positive brand image**.

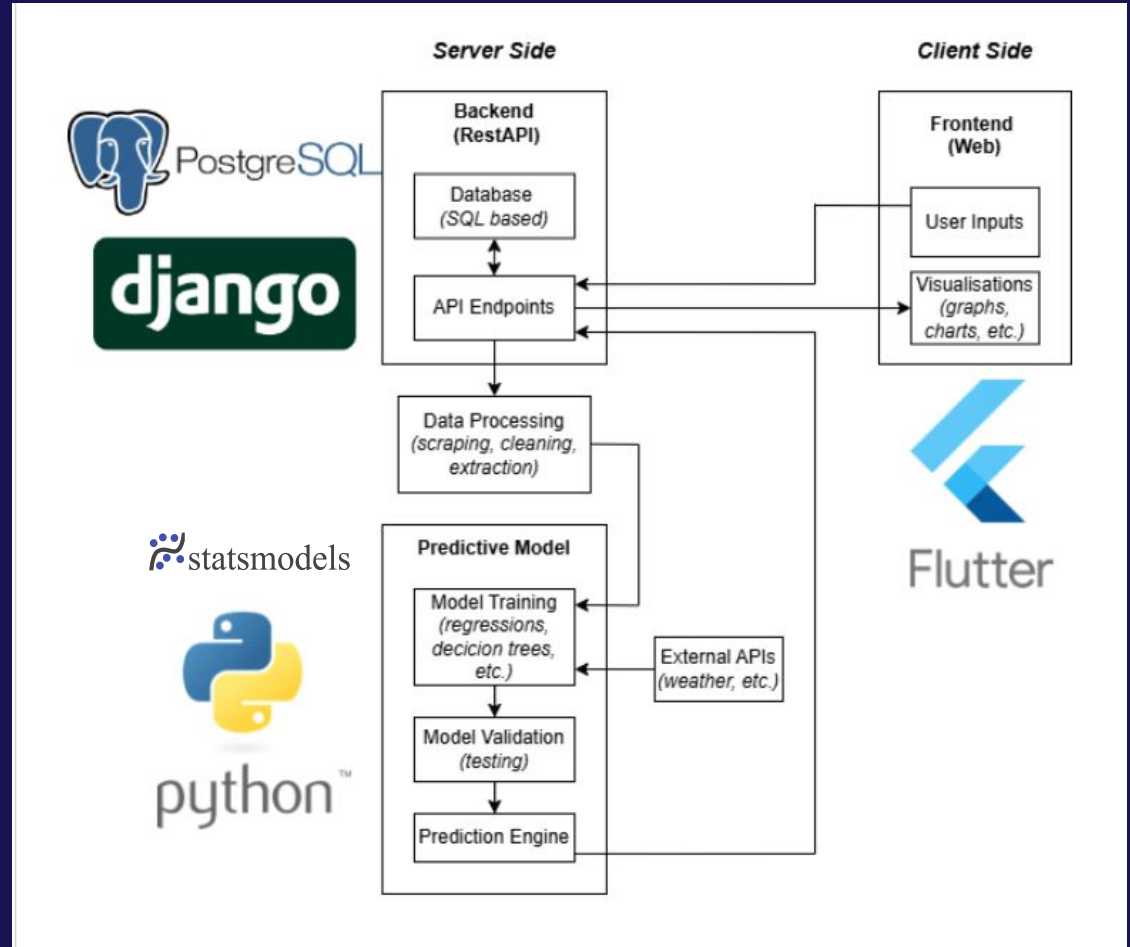


Railway Commuters

As a **railway commuter**, I want companies to adopt Synoptix technology to prevent incidents, so that I can feel **safer and more confident** when using the railway system.



Tech Stack + Architecture



Interface Design - 1

Railway Traffic Analysis

Please enter risk factor parameters into the fields below. We will then determine the associated risks based on these factors.

Amount of rain (mm):

Temperature (degrees C):

Proximity of crossing (km):

Train speed (mph):

Wind speed (mph):

Interface Design - 2

SYNOPTIX

SYNOPTIX

Railway Traffic Analysis

Train External Data

Type of weather
Clear

Precipitation
mm

Temperature
°C

Location

Time
Time

Train Internal Data

Type of Train

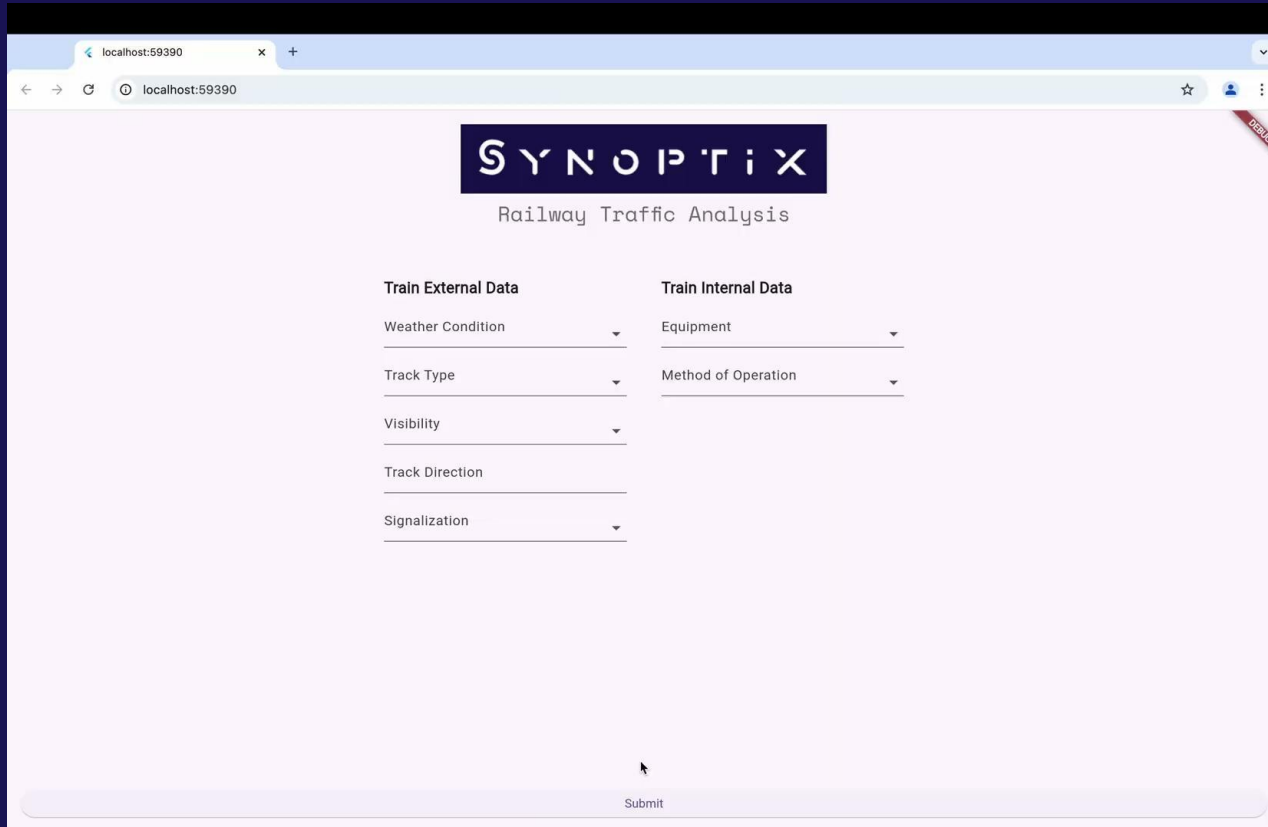
Train Speed
0-250 Mph

Accident in the last 5 years

Angle of the Train

Number of Signals

Product Demo (frontend)



localhost:59390

SYNOPSIS

Railway Traffic Analysis

Train External Data

Weather Condition

Track Type

Visibility

Track Direction

Signalization

Train Internal Data

Equipment

Method of Operation

Submit

Product Demo (current predictive model)

MNLogit Regression Results							
Dep. Variable:	y	No. Observations:	100				
Model:	MNLogit	Df Residuals:	-15				
Method:	MLE	Df Model:	110				
Date:	Tue, 12 Nov 2024	Pseudo R-squ.:	0.5265				
Time:	19:49:54	Log-Likelihood:	-79.586				
converged:	False	LL-Null:	-168.09				
Covariance Type:	nonrobust	LLR p-value:	5.347e-05				
	y=accident_type_Explosive-detonation	coef	std err	z	P> z	[0.025	0.975]
temperature		0.3426	0.173	1.978	0.048	0.003	0.682
track_class		-1.2701	0.728	-1.745	0.081	-2.696	0.156
track_density		0.0353	0.026	1.333	0.183	-0.017	0.087
equipment_type		-1.4062	0.745	-1.887	0.059	-2.867	0.054
equipment_attended		-13.1520	10.938	-1.202	0.229	-34.590	8.286
train_speed		0.0203	0.052	0.393	0.694	-0.081	0.122
remote_control_locomotive		-2.4196	1.934	-1.251	0.211	-6.209	1.370
positive_alcohol_tests		0.7272	0.456	1.596	0.110	-0.166	1.620
positive_drug_tests		-0.0542	0.231	-0.234	0.815	-0.508	0.399
visibility_dawn		0.3890	2.408	0.162	0.872	-4.332	5.110
visibility_day		-12.8727	7.824	-1.645	0.100	-28.208	2.462
visibility_dusk		-7.1932	4.673	-1.539	0.124	-16.353	1.966
weather_condition_cloudy		-3.4319	4.284	-0.801	0.423	-11.828	4.964
weather_condition_fog		4.5593	3.617	1.260	0.208	-2.531	11.649
weather_condition_rain		-8.4325	8.459	-0.997	0.319	-25.012	8.147
weather_condition_sleet		4.4000	3.872	1.136	0.256	-3.190	11.990
weather_condition_snow		-12.4009	8.452	-1.467	0.142	-28.966	4.165
track_type_main		-12.7713	7.401	-1.726	0.084	-27.278	1.735
track_type_siding		-7.3775	4.280	-1.724	0.085	-15.765	1.010
track_type_yard		-21.1195	10.990	-1.922	0.055	-42.660	0.421
method_of_operation_Direct Train Control		9.7192	6.285	1.546	0.122	-2.599	22.038
method_of_operation_Signal Indication		-2.9255	3.127	-0.936	0.350	-9.055	3.203
method_of_operation_Yard/Restricted Limits		17.2128	9.505	1.811	0.070	-1.416	35.842
	y=accident_type_Fire/violent rupture	coef	std err	z	P> z	[0.025	0.975]
temperature		0.0207	0.020	1.030	0.303	-0.019	0.060
track_class		0.0461	0.184	0.251	0.802	-0.406	0.314

```
← → ↺ ⓘ http://127.0.0.1:8000/view_model_result/

Pretty-print ☒

{
  "Derailment": {
    "0": 0
  },
  "Fire/violent rupture": {
    "0": 0.176298708
  },
  "Head on collision": {
    "0": 0.0245289262
  },
  "Hwy-rail crossing": {
    "0": 0.1168114646
  },
  "Obstruction": {
    "0": 0.0366203834
  },
  "Other (describe in narrative)": {
    "0": 0.0971297385
  },
  "Other impacts": {
    "0": 0.377528056
  },
  "RR grade crossing": {
    "0": 0.0240681937
  },
  "Raking collision": {
    "0": 0.0369422781
  },
  "Rear end collision": {
    "0": 0.0425723955
  },
  "Side collision": {
    "0": 0.067499856
  }
}
```

Issues We Faced

- Changing from React to Flutter
 - Miscommunication on WhatsApp due to flow of messages
 - Solution: relevant team members tagged in messages, ensure messages are reacted to
- Collaborative coding in Github:
 - Initialised the Flutter project multiple times
 - Pushed several different initialisations to different branches
 - Merge conflicts had to be manually resolved - helped each other do this

Issues We Faced

- Lack of detailed data
 - Client did not provide data
 - Struggled finding data from the UK that would allow us to implement what the client wanted.
 - Solution: used data from Federal Railroad Administration (U.S.A).
- Predictive model
 - Time spent researching statistical models that would fit our data

Future plans for product (model)

Meet the client for feedback
(happening this week!)

Improve model

Sanitize data!

Add more predictor
variables

Confidence intervals

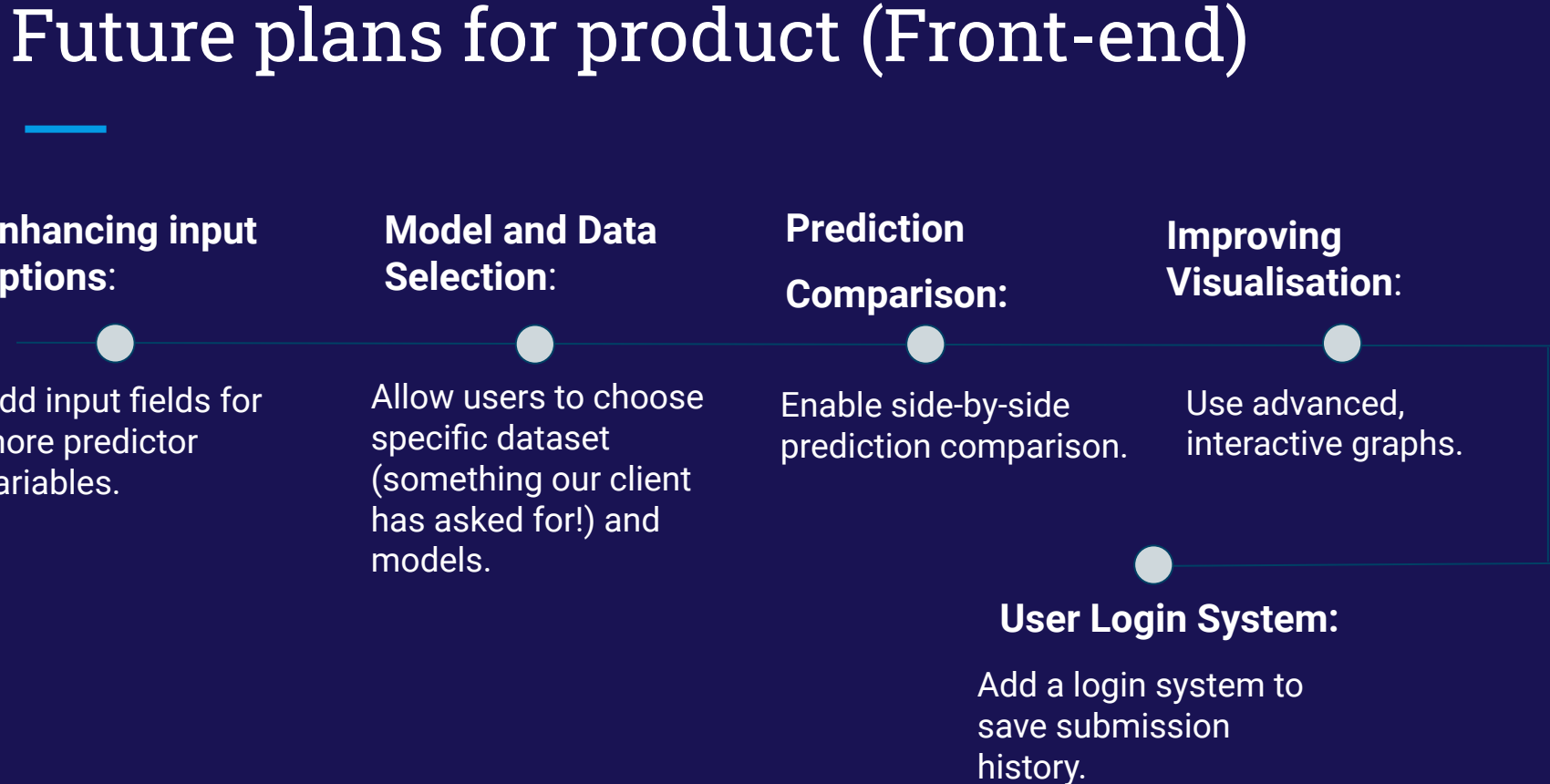
Improve performance

Experiment with different models

Use different
statistical models

Experiment with ML

Future plans for product (Front-end)



```
graph LR; A[Enhancing input options:] --- B[Model and Data Selection:]; B --- C[Prediction Comparison:]; C --- D[Improving Visualisation:]; D --- E[User Login System:];
```

Enhancing input options:

Add input fields for more predictor variables.

Model and Data Selection:

Allow users to choose specific dataset (something our client has asked for!) and models.

Prediction Comparison:

Enable side-by-side prediction comparison.

Improving Visualisation:

Use advanced, interactive graphs.

User Login System:

Add a login system to save submission history.

Future plans for project management

- Alternating project manager roles
- Test-driven development
- Continue + expand continuous integration
- Start using continuous deployment

Questions